MILITARY SPECIFICATION

BODY ASSEMBLY, TANK PERISCOPE: 10513620

This amendment forms a part of Military Specification MIL-B-60133(MU) dated 22 January 1965.

Paragraph 2.1, under DRAWINGS add:

"D7560124 Millidioptometer, 20X"

Paragraph 2.1, under STANDARDS add:

"MIL-STD-810 Environmental Test Methods"

Paragraph 3.3.2: Delete and substitute:

"3.3.2 Rigidity.- The catches shall be capable of securing the body assembly rigidly (no apparent separation) against the locating surfaces of head and mount with which it is intended to be used, when vibrated successively along 3 mutually perpendicular axes. Vibration shall be a simple harmonic motion, the frequency varying either lineraly or logarithmically from 5 to 55 to 5 cps in 5-minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64-inch (1/32-inch total excursion). Subsequent, the body assemly shall meet all the requirements herein."

Paragraph 3.6: Delete last sentence and substitute: "The incident line of sight shall also emerge parallel to a plane, defined by the 2-inch-long shoulder (generated by the machining of the two pads) adjacent to Valve Assembly 8201751, within 45 minutes of arc."

Paragraph 3.7: Delete paragraph and substitute:

"3.7 Image tilt.- The image of a target line originally perpendicular to a plane, defined by the 2-inch-long shoulder (generated by the machining of the two pads) adjacent to Valve Assembly 8201751, shall emerge perpendicular to the same plane within 45 minutes of arc.

Add the following new paragraphs:

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three body assemblies

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plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

* Paragraph 4.5.2.2: Delete and substitute:

"4.5.2.2 Rigidity.- Standard vibrating equipment shall be utilized for this test. The vibrating machining shall be equipped with means to secure a holding fixture as shown in Figure I. Attach and clamp the body assembly in the holding fixture by means of the catches (see Figure I). The test shall be performed in accordance with Method 514, Procedure IX, Part I of MIL-STD-810 except that the vibration condition in 3.3.2 and the conditions as specified herein shall apply. The body assembly under test shall be vibrated along the three mutually perpendicular axes while the body assembly is maintained immobilized in the holding fixture. The amplitude of the applied vibration shall be monitored at the extreme bottom portion of the body assembly as positioned in the fixture. During vibration, check rigidity of the body assembly in the holding fixture using a stroboscope to produce the optical effect of stopping the vibration motion to determine rigidity at the locating and mating surfaces. The body assembly shall remain secure against the holding fixture receiving surfaces to assure conformance with the requirements of 3.3.2. At the conclusion of this test the body assembly subjected to the examination specified in 4.5.2.2.1."

Add the following new paragraph:

"4.5.2.2.1 Examination (post vibratory exposure).- Subsequent to the rigidity (vibration) test, the body assembly shall be examined for any evidence of physical failure such as; a visual examination shall be made for casting failure, glass breakage, missing or damaged parts. A functional check of the catches shall be made to determine that the body assembly is clamped securely in the holding fixture, when the catches reach their apex

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and are fully closed. Body assemblies that have satisfactorily passed the aforementioned examinations shall then be tested for and meet all the other requirements of this specification."

Paragraph 4.5.5: Delete and substitute:

"4.5.5 Light deviation.- This test shall be performed utilizing an auxiliary telescope, containing a horizontal and vertical reticle line, mounted on a supporting device. The device shall be capable of holding the telescope and body assembly in a position permitting an incident line of sight to be directed from a calibrated target in a collimator projector, through the body assembly to the auxiliary telescope. Utilize the verticle surface of the 2-inch shoulder adjacent to the valve assembly to centrally position of the body assembly in the device with the machined top surfaces of the pads on the same plane with the line of sight through the telescope. The target pattern shall consist of a horizontal and vertical crossline with limit marks 45 minutes from the intersection of the crossline. Limit marks shall also be added in each quadrant, offset 45 minutes of arc from the horizontal crossline, to subsequently measure image tilt. The collimating projector shall be positioned in front of the body assembly with the center vertical line of the target in a plane parallel to a plumb line and on the same plane with the vertical surface of the 2-inch shoulder adjacent to the valve assembly. The auxiliary telescope reticle line shall be adjusted parallel to a plumb line. With the body assembly installed in the device, the line of sight shall be determined by viewing through the telescope and observing coincidence between the intersecting point of the auxiliary telescopes reticle and the corresponding intersecting point of the collimator target line. This coincidence must be maintained within the limits of the tolerance line of the collimator target to assure conformance with the requirement outlined in 3.6."

Paragraph 4.5.7: Delete and substitute:

"4.5.7 Spherical power.- Spherical power shall be checked in accordance with the set-up and operating instructions outlined on Dwg. D7560124. The test equipment shall conform to Drawing D7560124, except that the target (Resolving power chart) shall contain four line sets as shown in Figure I of MIL-O-13830. Spherical power at any point in the scanning area shall not exceed the tolerance (millidiopter) specified in 3.8, as measured on the millidiopter scale."

Paragraph 4.5.8: Delete and substitute:

"4.5.8 Astigmatism.- Perform this test utilizing the test procedure in 4.5.7. Astigmatism at any point in the scanning area shall not exceed the tolerance (millidiopter) specified in 3.9, as measured on the millidiopter scale."

Paragraph 6.2: Add: "(e) Requirement for first article testing."

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Project No. 6650-A059

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